# Heaven Andrei E. Fabunan

**Registered Electrical Engineer (REE)**

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## PROFILE

Registered Electrical Engineer (REE) with a strong foundation in electrical systems, troubleshooting, and technical analysis. Skilled in applying engineering principles to solve real-world problems, optimize system performance, and support safe and efficient operations. Motivated, fast-learning, and comfortable working with technical tools, documentation, and cross-functional teams. Passionate about continuous improvement and practical engineering solutions.

## PROFESSIONAL EXPERIENCE

**Lucalles Production,** *Editing Manager (Freelancer)* 05/2026 – Present

- Review and quality-check edited videos before publication to ensure accuracy, consistency, and compliance with channel requirements.
- Assign editing projects to editors based on priorities, deadlines, and workload.
- Monitor the progress of multiple projects and coordinate with editors to ensure timely completion.
- Communicate revisions and quality requirements to editors and resolve issues before publication.
- Coordinate content production across multiple YouTube channels while maintaining consistent quality and delivery schedules.

**Sumifru Philippines X RAL,** *Electrical Engineering Intern* 06/2025 – 07/2025
Los Baños, Laguna

- Conducted background research on pest detection methods and available sensing technologies to conceptualize an Arduino-based solution using multiple sensors.
- Designed an Arduino-based pest detection device for a banana farm to identify the presence of pests in real field conditions.
- Selected and integrated appropriate sensors and electronic components based on project requirements and feasibility.

## EDUCATION

**University of the Philippines – Los Baños,** 2021 - 2026
*Bachelor of Science in Electrical Engineering*

**Calamba City Science Integrated School** 2019 - 2021
*Science, Technology, Engineering, and Mathematics*

## PROJECTS

**Direct Current Motor (DC Motor)** 2023

- Built a small-scale DC motor from scratch to demonstrate electromagnetic interaction, mechanical rotation, and basic motor construction principles.

**Circuit Total Resistance Calculator Program** 2023

- Developed a desktop program that calculates total resistance for series and parallel resistor networks.
- Automated circuit calculations to enable faster evaluation of different resistor configurations.

**Device Charger** 2024

- Designed a regulated DC charger using a Zener diode circuit to produce a stable 8-V output.

**Audio Amplifier** 2024

- Constructed and tested an audio amplifier circuit to improve signal strength and sound clarity.
- Analyzed component behavior and amplifier gain to evaluate circuit performance.

**Shell Type Transformer** 2024

- Designed and evaluated a shell-type transformer to study magnetic flux paths, winding configuration, and voltage transformation under AC operation.

**Water Level Detection Device** 2024

- Developed an Arduino-based water level monitoring system using an ultrasonic sensor for real-time liquid-level measurement.
- Integrated sensor inputs and microcontroller programming for basic automation and monitoring.

**Magnitude Comparator with Digital Subsystems** 2024 - 2025

- Implemented a 2-bit magnitude comparator integrated with a full adder, MOD-13 up-counter, and custom LED pattern.
- Integrated multiple digital logic subsystems to demonstrate combined logic operation and circuit control.

**Self-Contained Pest Detection Device for Banana Farm** 2025

- Designed and assembled a field-ready pest detection device using a microcontroller and multiple sensors.
- Integrated sensing components to detect pest presence in an outdoor banana farm environment and support early monitoring and response.

**Transformer Design Program** 2025

- Developed a Visual Basic software tool for designing single-phase and three-phase power and distribution transformers.
- Implemented selectable core-type and shell-type configurations with automated dimensional and electrical calculations.
- Added cooling-method recommendations, default design generation, output export, and input error detection.

**Electrical System Design – Commercial Mall** 2025

- Designed the electrical system of a commercial mall using AutoCAD.
- Performed illumination calculations, load scheduling, voltage-drop analysis, feeder and panel sizing, and transformer sizing.
- Prepared electrical riser diagrams and coordinated major electrical system components.

**Thesis – Optimization of Vehicle-to-Grid Charging and Discharging Power Levels to Reduce Total Harmonic Distortion in an IEEE 37-Bus System using Grey Wolf Optimization** 2025

- Optimized V2G charging and discharging power levels in an IEEE 37-bus distribution system at 62% EV penetration to reduce voltage distortion.
- Developed and evaluated the optimization using MATLAB and OpenDSS.
- Achieved average bus voltage THD below the 5% IEEE 519 target while improving voltage profiles.

## RESEARCH PRESENTATION

**International Research Presentation – ISAT-25** 2026

- Presented research on optimizing Vehicle-to-Grid (V2G) charging and discharging power levels to reduce Total Harmonic Distortion (THD) in an IEEE 37-Bus System.
- Presented at the 25th International Symposium on Advanced Technology (ISAT-25), Southern Taiwan University of Science and Technology (STUST), Tainan, Taiwan.

## SKILLS

**Power Systems & Power Quality**

- V2G power control and optimization
- THD analysis and harmonic mitigation
- Voltage profile and power-quality assessment (aligned with IEEE standards)

**Programming & Engineering Tools**

- Distribution system simulation (OpenDSS)
- Numerical modeling and optimization in MATLAB
- Metaheuristic optimization (Grey Wolf Optimization – GWO)
- Visual Basic for engineering calculation and design tools

**Embedded Systems**

- Microcontroller programming and sensor interfacing using Arduino

**Electronics & Digital Systems**

- Analog circuit design (amplifiers, regulators)
- Digital logic design (comparators, counters, adders)
- DC machines and transformer fundamentals

**Electrical Building Systems**

- Load scheduling, voltage-drop analysis, feeder and panel sizing
- Transformer sizing and riser diagram preparation

**CAD & Drafting**

- Electrical and technical drafting using AutoCAD
- Manual technical drafting